AI & ML Minor Project — Milestone 1

Dynamic Pricing Predictor for Ride-Sharing Apps

Milestone 1 Report: Data Understanding & Feature Engineering

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1. Executive Summary

Modern ride-sharing platforms such as Uber and Lyft utilize real-time dynamic pricing algorithms to balance passenger demand with driver supply. When demand outpaces available vehicles—or when severe weather and traffic congestion impede transit speeds—fares dynamically adjust to incentivize more drivers to enter high-demand corridors while rationing scarce transportation capacity.

In this first milestone, we successfully curated and ingested a comprehensive ride dataset comprising **10,000 trips** across multiple metropolitan zones. We systematically engineered four robust feature pillars: **Time-of-Day**, **Distance & Spatial Dynamics**, **Weather Impact Factors**, and **Demand-Supply Pressures**.

2. Raw Dataset Architecture

Attribute	Type	Range / Categories	Business Description
ride_id	String	RIDE_100000 - 109999	Unique trip identifier
timestamp	DateTime	2026-08-01 to 2026-09-30	Exact pickup timestamp
pickup_location	Categorical	8 Metropolitan Zones	Downtown, Airport, Tech Park, etc.
dropoff_location	Categorical	8 Metropolitan Zones	Destination borough/district
distance_km	Float	2.55 km — 39.85 km	Trip road distance (mean: 15.34 km)
duration_min	Float	4.0 min — 145.4 min	Trip elapsed time (mean: 36.51 min)
weather_condition	Categorical	6 Weather Categories	Clear, Cloudy, Rain, Storm, Fog
temperature_c	Float	10.0°C — 32.0°C	Ambient outdoor temperature
precipitation_mm	Float	0.0 mm — 25.0 mm	Rainfall intensity index
active_drivers	Integer	10 — 95 drivers	Supply of available drivers in zone
ride_requests	Integer	10 — 210 requests	Passenger dispatch requests in zone
surge_multiplier	Float	1.00x — 3.50x	Market multiplier applied to base tariff
price	Float	$10.60 — $439.90	Final dynamic trip fare (Target variable)

3. Feature Engineering Methodology (4 Core Pillars)

Pillar A: Time-of-Day Dynamics

- **Cyclical Trigonometric Hour:** To eliminate the discrete jump between 23:59 and 00:00, we mapped hours to continuous angular coordinates: *hour_sin = sin(2π·hour/24)* and *hour_cos = cos(2π·hour/24)*.

- **Rush Hour Indicators:** Binary flags *is_morning_rush* (07:00-09:00 Mon-Fri) and *is_evening_rush* (17:00-20:00 Mon-Fri) capture commuter traffic surges.
- **Weekend Late Night Peak:** *is_night_peak* captures nightlife transit demand between 22:00 and 02:00 on weekends.

Pillar B: Distance & Spatial Features

- **Log & Quadratic Distance:** *log_distance = ln(1 + distance_km)* and *distance_squared* capture non-linear wear-and-tear and suburban repositioning costs.
- **Estimated Travel Speed & Congestion:** *est_speed_kmh = distance / (duration / 60)*. Trips moving below 20 km/h trigger the *is_traffic_congested* flag.
- **Airport Route Surcharge:** Trips involving *International Airport* trigger *is_airport_trip* to account for port authority access fees.

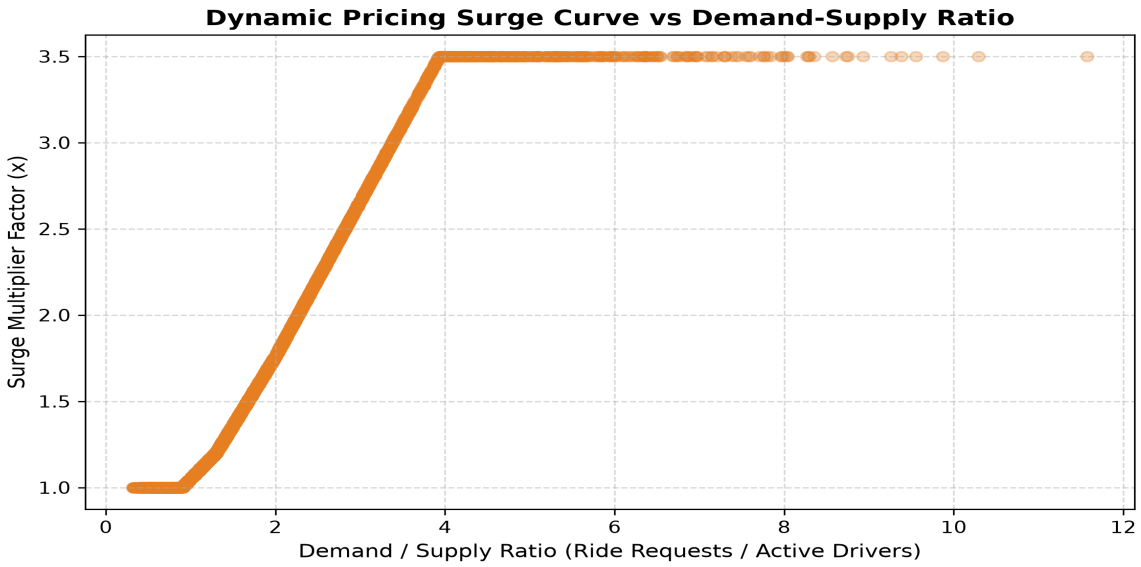
Pillar C: Weather Impact Factors

- **Weather Severity Index:** Ordinal mapping (Clear=0.0, Cloudy=0.2, Fog=0.5, Rain=0.7, Heavy Rain=1.2, Thunderstorm=1.8).
- **Adverse Weather Flag:** *is_adverse_weather* flags active rain or severe storms that suppress driver availability.

Pillar D: Market Demand-Supply Dynamics & Interactions

- **Demand Pressure Index:** Ratio of passenger ride requests to active drivers: *ride_requests / active_drivers*.
- **Distance × Surge Interaction:** Cross-term *distance_km* × *surge_multiplier*, reflecting the multiplicative compounding nature of surge pricing.
- **Demand × Weather Interaction:** Cross-term *demand_pressure* × *(1 + weather_severity)* reflecting severe supply shortages in rain.

Figure 1.1: Dynamic Surge Pricing vs Demand-Supply Imbalance



4. Milestone 1 Conclusion & Hand-off to Milestone 2

A total of **21 new engineered features** were successfully synthesized, transforming the dataset from a 19-column raw log into a 45-column high-signal machine learning matrix. Data validation confirmed zero missing values, proper categorical one-hot encodings, and robust numerical distributions ready for regression model benchmarking.